

# DS Assignment 1 — Arrays, Linked Lists & Complexity

Data Structures (DS) · Unit 1 · Arrays, Linked Lists & Complexity · 2025

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Answer all questions. Show working and cite references where used.

## Questions

1. Write short notes on Asymptotic Notation.
2. Discuss the role of Arrays in Arrays, Linked Lists & Complexity.
3. Explain Singly Linked Lists with a neat diagram and a suitable example.
4. Define Doubly and Circular Linked Lists. Derive or justify the key properties involved.
5. Discuss the role of Sparse Matrices in Arrays, Linked Lists & Complexity.
6. What is meant by Polynomial Representation? Discuss its advantages and limitations.